

A Comparative Analysis of Energy Efficiency Across Large Language Models

Fatin Israq Talha, Mehrab Sadekin Borno, Md Rakib Hasan Rabbi, Muhmmad Ahamadul Hasan Prianto

Department of Computer Science and Engineering
United International University, Dhaka, Bangladesh

Abstract

The rapid adoption of large language models (LLMs) has led to a common but energy-inefficient practice: routing all user prompts, regardless of complexity, through the same large-scale model. This uniform approach incurs substantial and often unnecessary energy costs, as many prompts (e.g., simple factual queries or short transformations) do not require the full capacity of a large model to be answered accurately. In this work, we first conduct an empirical study measuring the energy consumption and output quality of multiple open-source language models across varying sizes and quantization levels, using real-time GPU power measurements. Building on these measurements, we propose a lightweight prompt-complexity router that dynamically selects the smallest capable model for a given input, reserving larger models only for prompts that genuinely require them. We evaluate our system in terms of total energy consumption, response quality, and latency, comparing it against a baseline that always uses a single large model. Our results demonstrate that adaptive model routing can substantially reduce energy consumption with minimal degradation in answer quality, offering a practical and generalizable strategy for building more sustainable AI-serving systems.

Keywords: Large Language Models, Energy Efficiency, Green Computing, Model Quantization, Model Routing, Sustainable AI

1 Introduction

The rapid growth of Large Language Models (LLMs) has significantly increased the computational and energy costs of AI inference. A common serving strategy is to route every user query to the same large model, even though many queries are simple and can be handled by smaller models. This results in unnecessary energy consumption and computational resource usage. Recent research has explored several approaches to improve LLM efficiency, including energy measurement, model quantization, and adaptive model routing. However, these approaches are often studied independently. In particular, existing routing methods primarily focus on cost, latency, or answer quality, while energy consumption is rarely measured directly using real-time GPU power data. To address this gap, this study investigates the energy efficiency of LLMs across different model sizes and quantization levels. Based on these measurements, we propose a lightweight prompt-complexity router that dynamically selects the smallest model capable of handling each query. Larger models are used only when necessary. The proposed approach is evaluated in terms of energy consumption, response quality, and latency, with the goal of reducing the environmental and computational cost of LLM inference while maintaining acceptable answer quality.

2 Related Works

2.1 Literature Review

The growing energy footprint of large language model (LLM) inference has motivated a substantial body of work that can be organized into four broad, interconnected strands: (i) empirical measurement of LLM inference energy, (ii) quantization as a means of reducing the energy-accuracy trade-off, (iii) model routing and cascading strategies that avoid invoking large models unnecessarily, and (iv) foundational work establishing the carbon and environmental footprint of deep learning and motivating the broader Green AI agenda.

A first line of work focuses on directly profiling the energy consumed during LLM inference under real hardware conditions. Samsi et al. [1] benchmark the energy costs of several open-source LLMs on GPU infrastructure, establishing baseline power and latency figures that later routing and quantization studies build upon. Fernandez et al. [2] extend this measurement effort by explicitly considering efficiency optimizations, framing energy consumption as a first-class metric alongside accuracy and latency. Complementing these single-node measurement studies, Wilkins et al. [3] show that heterogeneous GPU clusters can be scheduled to lower the aggregate energy consumption of inference workloads, suggesting that energy savings can be achieved not only at the model level but also at the infrastructure-scheduling level. Ji and Jiang [4] provide a systematic review of electricity demand associated with LLMs, consolidating evaluation methodologies and open challenges across the field. Most recently, Niu et al. [5] introduce TokenPowerBench, a benchmark that captures GPU-, node-, and system-level power draw and attributes energy separately to the prefill and decode phases of inference, offering fine-grained evidence that informs the design of measurement pipelines such as the one used in this project.

A second strand investigates quantization as a direct lever for reducing the energy cost of inference while attempting to preserve output quality. Frantar et al. [6] introduce GPTQ, a post-training quantization method that compresses model weights with minimal accuracy loss, and Lin et al. [7] propose AWQ, which protects a small subset of salient weights to further improve the accuracy-compression trade-off. Xiao et al. [8] address the difficulty of quantizing activations by smoothing outlier values prior to quantization, and Dettmers et al. [9] demonstrate that 8-bit matrix multiplication can match full-precision performance at large model scale. Collectively, this strand establishes that quantization level is a controllable variable that trades a measurable amount of accuracy for a measurable amount of energy savings – precisely the trade-off this project seeks to characterize empirically.

A third strand, closely aligned with the routing component of this project, explores directing individual prompts to different models based on estimated complexity, rather than applying quantization uniformly. Chen et al. [10] propose FrugalGPT, a cascading strategy that queries progressively larger LLMs only when earlier, cheaper models are deemed insufficiently confident, substantially reducing cost while retaining accuracy. Ong et al. [11] learn a routing function from human-preference data to decide, per query, whether a smaller or larger model should respond. Ding et al. [12] formalize a similar problem as cost-efficient, quality-aware query routing between a small and a large model. Khan et al. [13] present a case study connecting routing-style optimizations to energy-efficiency metrics specifically, rather than cost alone. While these works demonstrate that routing can reduce inference cost or latency, comparatively few directly measure the resulting energy savings using real-time hardware power measurements, which is a gap this project’s proposed router is designed to address.

Finally, a foundational strand of work established the environmental case for energy-aware AI system design in the first place. Strubell et al. [14] were among the first to quantify the carbon emissions of training large NLP models, prompting widespread attention to the sustainability of deep learning research. Schwartz et al. [15] formalize the notion of “Green AI,” arguing for efficiency as an explicit research objective alongside accuracy. Li et al. [16] draw attention to the often-overlooked water footprint of AI model training and serving. Luccioni et al. [17] provide a detailed carbon-footprint case study of the 176B-parameter BLOOM model, illustrating the scale of emissions associated with large-model deployment. Together, these works motivate the broader problem this project addresses: reducing the operational energy footprint of LLM inference through model-level and system-level design choices.

2.2 Gap Analysis

Table 1 summarizes the reviewed literature according to problem definition, contribution, methodology, and the specific gap that motivates the present work. Across the reviewed studies, three recurring gaps emerge. First, energy-measurement studies [1–5] establish sound measurement methodology but rarely combine those measurements with an actionable serving-time decision mechanism. Second, quantization studies [6–9] characterize the energy-accuracy trade-off of compressing a single model but do not consider dynamically choosing among multiple models of different sizes per prompt. Third, routing and cascading studies [10–13] optimize primarily for monetary cost or latency, with energy treated as, at best, a secondary or estimated metric rather than one measured directly from GPU power draw. This project is positioned to close these gaps by combining direct, real-time GPU energy measurement (as in the first strand) with a prompt-complexity router (as in the third strand) that additionally accounts for quantization level (as in the second strand) when selecting a model, and by evaluating the resulting system explicitly on measured energy consumption rather than proxy cost metrics.

As summarized above, no reviewed work simultaneously (1) measures LLM inference energy directly via real-time GPU power draw, (2) incorporates model quantization level as a routing variable, and (3) makes a per-prompt, complexity-aware model-selection decision at serving time. This combined gap is the primary motivation for the energy-aware prompt-complexity router proposed in this work.

Table 1: Gap analysis of the papers most closely related to this project.

Ref.	Problem Definition	Contribution	Methodology	Gap Found
[1]	There is no common benchmark for how much energy popular LLMs actually use.	Benchmarked the energy cost of several open-source LLMs on GPUs.	Measured GPU power draw while models processed a fixed set of prompts.	Gives useful energy numbers but doesn't decide which model to use for a given task.
[5]	Existing energy tools focus on training or overall inference performance, without a lightweight way to attribute power specifically within inference.	TokenPowerBench: an open-source benchmark measuring GPU-, node-, and system-level power, split by prefill and decode phase.	Ran multiple open-source model families (up to 405B params) under varying batch size, context length, and quantization; measured joules/token.	Provides a reusable, phase-aware energy-measurement framework, but doesn't use the measurements to pick a model or quantization level per prompt.
[10]	Sending every question to a large, expensive model wastes money.	FrugalGPT: try cheaper models first, only escalate to a bigger model if needed.	Cascades models and checks confidence before moving to a bigger one.	Built to save money and keep accuracy up – not built around measured energy use.
[11]	Need an automatic way to decide which model should answer a given query.	RouteLLM: a router trained on human-preference data to pick small vs. large models.	Trains a classifier that routes each query to a small or large model.	Optimized for answer quality and cost, not for measured energy savings.
[13]	Many LLM optimization studies don't clearly connect their techniques to energy savings.	A case study linking several optimization techniques to energy-efficiency outcomes.	Evaluated existing optimization techniques against energy and performance metrics.	Looks back at existing techniques rather than building a live system that routes prompts to save energy.

3 Methodology

This section outlines the experimental methodology designed to evaluate the energy-quality trade-offs of adaptive LLM serving. The architecture comprises an empirical energy-profiling harness across heterogeneous consumer GPUs, a tiered model candidate pool, a lightweight routing mechanism, and a multi-dimensional evaluation protocol.

3.1 Experimental Hardware and Measurement Protocol

To reflect practical deployment environments and analyze architectural power scaling across generations, experiments are conducted on two dedicated consumer GPU platforms sharing an 8 GB VRAM budget:

- **Platform A:** NVIDIA GeForce RTX 4060 (8 GB GDDR6, Ada Lovelace architecture, 115 W TDP, TSMC 4N process).
- **Platform B:** NVIDIA GeForce RTX 3060 Ti (8 GB GDDR6, Ampere architecture, 200 W TDP, Samsung 8nm process).

Inference is executed locally using quantized GGUF format weights served via high-throughput local execution engines. Dynamic power draw $P(t)$ is continuously sampled at 50 ms intervals using the NVIDIA Management Library (NVML/`pynvml`) via an isolated background daemon to avoid inference contention. Total energy consumption E (in Joules) for a request executing over duration $[0, T]$ is numerically integrated using the trapezoidal rule:

$$E = \sum_{k=1}^{N-1} \frac{P(t_k) + P(t_{k+1})}{2} \cdot (t_{k+1} - t_k) \quad (1)$$

Static baseline/idle power draw (P_{idle}) is profiled prior to each batch and subtracted to compute net dynamic inference energy (E_{net}).

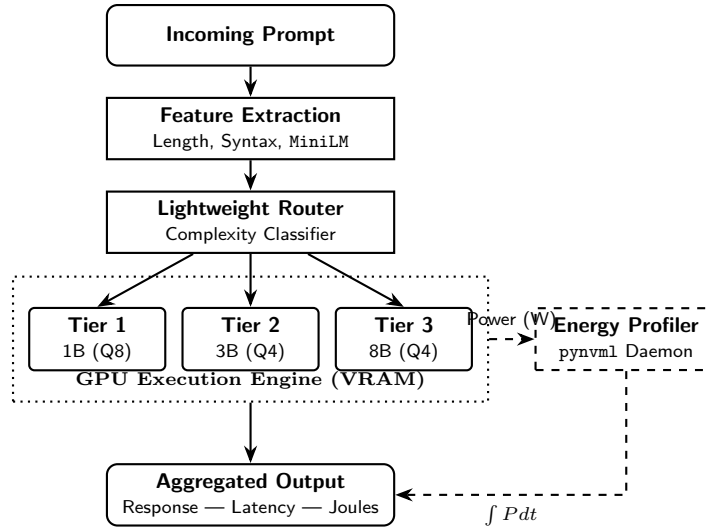


Figure 1: System architecture pipeline demonstrating the adaptive routing mechanism and real-time energy profiling.

3.2 Model Candidate Pool and Quantization Tiers

Given the strict 8 GB memory boundary, the model candidate pool is organized into three operational capability tiers, varying both parameter scale and quantization precision:

- **Tier 1 (Lightweight / Low Complexity):** Llama-3.2-1B-Instruct (8-bit quantization, Q8_0, ~1.3 GB VRAM footprint). Target tasks include binary sentiment classification, intent detection, and short factual extraction.
- **Tier 2 (Intermediate / Medium Complexity):** Llama-3.2-3B-Instruct / Phi-3.5-mini-3.8B (4-bit quantization, Q4_K_M, ~2.6–3.1 GB VRAM footprint). Target tasks include multi-sentence summarization, JSON transformation, and single-step queries.
- **Tier 3 (Heavyweight / High Complexity):** Llama-3.1-8B-Instruct / Mistral-7B-Instruct-v0.3 (4-bit quantization, Q4_K_M, ~5.2–5.8 GB VRAM footprint). Target tasks include multi-step mathematical reasoning, code synthesis, and complex contextual deduction.

3.3 Lightweight Prompt-Complexity Router Design

The router must determine the minimal adequate model tier per query while ensuring that routing overhead (Δt_{route} , ΔE_{route}) remains negligible compared to inference costs. We implement a two-stage lightweight decision pipeline:

1. **Feature Extraction:** Extracts surface-level heuristics (token length, character count, code syntax markers like `def`, `class`, `curl`) concatenated with a dense semantic sentence embedding generated by `all-MiniLM-L6-v2` (22M parameters) executed on host CPU.
2. **Tier Classification:** A lightweight multi-class classifier (Logistic Regression with L_2 regularization / Random Forest) maps the feature vector to one of the target tiers $\hat{y} \in \{\text{Tier 1, Tier 2, Tier 3}\}$.

3.4 Benchmark Dataset and Workload Distribution

The system is evaluated against a curated composite workload of 600 prompts (200 prompts per complexity class):

- **Low-Complexity Subset:** SST-2 (sentiment classification) and SQuAD v2.0 (closed-domain fact retrieval).
- **Medium-Complexity Subset:** CNN/DailyMail (abstractive summarization) and Multi-genre Natural Language Inference (MNLI).
- **High-Complexity Subset:** GSM8K (multi-step grade-school mathematics) and HumanEval (Python functional synthesis).

3.5 Evaluation Metrics

System performance is assessed across three primary dimensions:

1. **Energy Efficiency:** Total electrical energy consumed per workload (E_{total} in kJ), energy per output token (Joules/Token = E/N_{tokens}), and microarchitectural energy scaling ratio ($\eta = E_{\text{RTX 3060 Ti}}/E_{\text{RTX 4060}}$).
2. **Quality Retention:** Output correctness evaluated against ground truth using Accuracy/Exact Match (EM) for classification/QA, ROUGE-L for summarization, and Pass@1 for code generation. We compute the Quality Preservation Ratio (QPR) relative to the static 8B baseline:

$$\text{QPR} = \frac{\text{Score}_{\text{Router}}}{\text{Score}_{\text{Tier 3 Static}}} \times 100\% \quad (2)$$

3. **Latency Profile:** Time-to-First-Token (TTFT), End-to-End Latency, and Router Overhead (Δt_{router}).

References

- [1] S. Samsi, D. Zhao, J. McDonald, B. Li, A. Michaleas, M. Jones, W. Bergeron, J. Kepner, D. Tiwari, and V. Gadepally, “From words to watts: Benchmarking the energy costs of large language model inference,” in *2023 IEEE High Performance Extreme Computing Conference (HPEC)*, 2023, pp. 1–9.
- [2] J. Fernandez, C. Na, V. Tiwari, Y. Bisk, S. Luccioni, and E. Strubell, “Energy considerations of large language model inference and efficiency optimizations,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Vienna, Austria, 2025, pp. 32 556–32 569.
- [3] G. Wilkins, S. Keshav, and R. Mortier, “Hybrid heterogeneous clusters can lower the energy consumption of LLM inference workloads,” in *Proceedings of the 15th ACM International Conference on Future and Sustainable Energy Systems (e-Energy ’24)*, 2024, pp. 506–513.
- [4] Z. Ji and M. Jiang, “A systematic review of electricity demand for large language models: Evaluations, challenges, and solutions,” *Renewable and Sustainable Energy Reviews*, 2025.
- [5] C. Niu, W. Zhang, J. Li, Y. Zhao, T. Wang, X. Wang, and Y. Chen, “TokenPowerBench: Benchmarking the power consumption of LLM inference,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 38, 2026, pp. 32 582–32 590.
- [6] E. Frantar, S. Ashkboos, T. Hoefler, and D. Alistarh, “OPTQ: Accurate quantization for generative pre-trained transformers,” in *Proceedings of the 11th International Conference on Learning Representations (ICLR)*, 2023.
- [7] J. Lin, J. Tang, H. Tang, S. Yang, W.-M. Chen, W.-C. Wang, G. Xiao, X. Dang, C. Gan, and S. Han, “AWQ: Activation-aware weight quantization for LLM compression and acceleration,” in *Proceedings of Machine Learning and Systems*, vol. 6, 2024, pp. 87–100.
- [8] G. Xiao, J. Lin, M. Seznec, H. Wu, J. Demouth, and S. Han, “SmoothQuant: Accurate and efficient post-training quantization for large language models,” in *International Conference on Machine Learning (ICML)*, 2023, pp. 38 087–38 099.
- [9] T. Dettmers, M. Lewis, Y. Belkada, and L. Zettlemoyer, “LLM.int8(): 8-bit matrix multiplication for transformers at scale,” in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 30 318–30 332.
- [10] L. Chen, M. Zaharia, and J. Zou, “FrugalGPT: How to use large language models while reducing cost and improving performance,” *Transactions on Machine Learning Research*, 2024.
- [11] I. Ong, A. Almahairi, V. Wu, W.-L. Chiang, T. Wu, J. E. Gonzalez, M. Kadous, and I. Stoica, “RouteLLM: Learning to route LLMs with preference data,” in *Proceedings of the 13th International Conference on Learning Representations (ICLR)*, 2025.
- [12] D. Ding, A. Mallick, C. Wang, R. Sim, S. Mukherjee, V. Ruhle, L. V. S. Lakshmanan, and A. H. Awadallah, “Hybrid LLM: Cost-efficient and quality-aware query routing,” in *International Conference on Learning Representations (ICLR)*, 2024.

- [13] T. Khan, S. Motie, S. A. Kocak, and S. Raza, “Optimizing large language models: Metrics, energy efficiency, and case study insights,” in *2025 IEEE Conference on Artificial Intelligence (CAI)*, 2025, pp. 370–375.
- [14] E. Strubell, A. Ganesh, and A. McCallum, “Energy and policy considerations for deep learning in NLP,” in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, Florence, Italy, 2019, pp. 3645–3650.
- [15] R. Schwartz, J. Dodge, N. A. Smith, and O. Etzioni, “Green AI,” *Communications of the ACM*, vol. 63, no. 12, pp. 54–63, 2020.
- [16] P. Li, J. Yang, M. A. Islam, and S. Ren, “Making AI less “thirsty”: Uncovering and addressing the secret water footprint of AI models,” *Communications of the ACM*, vol. 68, no. 7, pp. 54–61, 2025.
- [17] A. S. Luccioni, S. Viguier, and A.-L. Ligozat, “Estimating the carbon footprint of BLOOM, a 176b parameter language model,” *Journal of Machine Learning Research*, vol. 24, no. 253, pp. 1–15, 2023.